RAG Quick Guide

A beginner's guide to the document question-answering pipeline.

What is RAG?

Retrieval-Augmented Generation combines document search with language-model generation. The system retrieves evidence and supplies it to the model as context.

Step 1 — Read data

Extract readable text from PDF, DOCX, and TXT files. PDF files that contain scanned images need OCR, which is outside this demo.

Step 2 — Chunking

Split long text into smaller overlapping passages. The example application uses a target chunk size of about 900 characters and an overlap of about 150 characters.

Step 3 — Indexing

ChromaDB stores the text chunks and their embeddings. The app also stores metadata including the source filename and chunk number.

Step 4 — Ask a question

A student enters a question. The application can either search the indexed chunks or skip search.

Step 5 — YES or NO

YES: retrieve the most relevant chunks and send them with the question to Claude. NO: send the question without retrieved document context. The NO path is useful as a baseline comparison.

Example

Question: Why do we chunk documents? A good answer explains that smaller passages make retrieval more focused and reduce the amount of context that must be passed to the model.

Remember

Retrieved text can be incomplete or irrelevant. A vector search result is evidence to inspect, not a guarantee that the answer is correct.